DEEPAK JHA

Associate Architect, Apra Labs | 10 Years | Full-Stack Systems & Agentic AI

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10 years building distributed real-time systems at scale — orchestrating **8,000+ concurrent devices** across multi-tenant, multi-region deployments with **zero-downtime failovers**. Authored product specs, defined features from scratch, and led cross-functional teams from requirements through production delivery. Building for the web since 2014 — shipped **15+ products** (4 Play Store apps, 11 npm tools for AI coding agents) with **25K+ users** and 11 open-source repos, as an early adopter of **agent-driven development**.

EXPERIENCE

Apra Labs — Associate Architect (2024–Present) · Senior SWE (2022–2024) · SWE (2019–2022) · **AlienHu** — Software Engineer (2017–2019)

SKILLS

LANGUAGES
C#
JavaScript / TypeScript
Python
Java
C / C++
PHP
Dart

FRONTEND
React / Redux
Angular
HTML5 Canvas
Electron

DATABASES
MySQL
MongoDB
Redis
CouchDB / PouchDB
Neo4j GraphDB

CLOUD & INFRA
Azure
AWS
GCP
Nginx

MOBILE
Flutter
Android
React Native
Ionic

REALTIME & AI
WebSockets / SignalR
WebRTC
MQTT / IoT
Machine Learning
NLTK

MORE
Node.js / Express
Agentic AI / MCP
Docker
CI / CD
Microservices
GraphQL

EDUCATION

B.E. in Computer Science & Engineering
SDMCET (2017), Karnataka

KEY PROJECTS

Video Surveillance System
2020 – Present

APRA LABS · ASSOCIATE ARCHITECT / LEAD

Architected an end-to-end cloud platform for orchestrating stable, **low-latency live streams**, configuration changes, and event captures for thousands of **geo-distributed cameras** across multiple tenants and regions. Engineered a scalable middleware layer with **localized brokers and edge protocols** for camera interfacing. Implemented reliable cross-region message routing via a **distributed service bus** and engineered **real-time signaling and streaming state management** across cloud boundaries for multi-region tenant isolation and seamless failovers. Led and mentored a cross-functional team — developers, QA, and support — driving stakeholder alignment across engineering, operations, and client teams from requirements through multi-region production rollouts.

Impact: Achieved **zero-downtime regional failovers** and sub-second stream switching, scaling to **8,000+ concurrent cameras** while onboarding new enterprise tenants without additional infrastructure investment

Tech stack: C# (.NET), Azure Service Bus, SignalR, Azure Functions, Redis, WebRTC, MQTT, ONVIF, Milestone VMS, React, Redux, NSIS

Real-Time Multipurpose Dashboard
2023 – Present

APRA LABS · ASSOCIATE ARCHITECT / LEAD

As lead engineer, mentored a junior developer while designing and building a modular, content-heavy dashboard system **from scratch** — rendering hundreds of **live-updating widgets** and **millions of concurrent data points** on a complex, cross-building timeline without blocking the UI main thread. Engineered a **custom HTML5 Canvas rendering pipeline** bypassing traditional DOM limitations. Offloaded heavy calculations, stream parsing, and data manipulation to background **Web Workers**. Applied **progressive data fetching**, localized slice queries, and **lazy evaluation** of off-screen states to minimize memory overhead.

Impact: Sustained a **locked 60 FPS** across millions of live data points with **zero browser memory leaks** — powering 24/7 facility operations across multi-building security and access-control environments

Tech stack: C#, SignalR, SQL, Stored Procedures, React, Redux, HTML5 Canvas, Web Workers, TypeScript

E-Commerce Middleware & POS Platform
2017 – 2019

ALIENHU · SOFTWARE ENGINEER

Designed and developed a full-stack **B2B point-of-sale application** with a team, and **single-handedly built** the end-to-end **B2C e-commerce mobile app** and the shared sync middleware managing inventory, order notifications, and payment flows across **fragmented POS networks**. Engineered a fault-tolerant **transaction queue with offline state management**, eliminating double-selling and inventory conflicts with **zero data loss** during resync.

Tech: Node.js, CouchDB/PouchDB, Angular, Electron, React Native, Ionic, WebSocket

OPEN SOURCE & SHIPPED PRODUCTS

ToggIDev
Toggle Android Developer Options instantly — widget, Quick Settings tile, Banking App Shield, Remote Access Control, Automation, and more. 4.5★ rated · 10K+ downloads.
Kotlin, Android

Terminal Studio
GPU-accelerated terminal multiplexer with built-in file browser, git diff viewer, code editor, and workspace persistence. Cross-platform, 120+ features, 15 themes.
Rust, wgpu, egui

Claude Shift
Align Claude Max 5-hour usage windows with your work schedule by scheduling strategic session pings during downtime. Cross-platform native OS schedulers.
TypeScript, Node.js

Claude Lifecycle Core
Middleware hub for Claude Code plugins — register once, dispatch all lifecycle events to subscriber plugins. Eliminates per-plugin hooks.json boilerplate.
Node.js, JavaScript

AGENTIC AI TOOLS: GhostPM · ToggIDev MCP · Claude Comedy · AntiBluff · AntiClimax · Ingred · SplitGame · Claude Ticker

ALSO: AutoDeck, CloseLink, AfterCall Shield (Kotlin) · GitReg (Rust) · tTag (Go) · Workspace ID Pro (VS Code)